5.7 Evidence and Result Presentation

Description and Priority: Presents the verification outcome and supporting evidence in a form understandable to the user. Priority: High.

Stimulus/Response Sequences:

1. Verification completes.
2. System prepares claim-level and response-level results.
3. Sources and evidence are attached.
4. The final result is displayed or returned to the requesting application.

Functional Requirements:

FR-23: The system shall display or return the overall verification result.

FR-24: The system shall display or return supporting evidence and source information for verified claims where available.

FR-25: The system shall distinguish supported, contradicted, and uncertain claims in the result.

5.8 Verification API

Description and Priority: Provides an interface through which another AI application can send responses to the module and receive verification results. Priority: High.

Stimulus/Response Sequences:

5. Developer integrates the API.
6. AI application sends a verification request.
7. Module processes the request.
8. API returns the verification result in the defined format.

Functional Requirements:

FR-26: The system shall provide an API endpoint or equivalent interface for response verification.

FR-27: The API shall return a structured verification result containing status and relevant evidence information.

FR-28: The API shall return an appropriate error response for invalid requests or unavailable verification services.

5.9 Trusted Source Management

Description and Priority: Allows authorized administrators to add, remove, enable, disable, or update the trusted sources used by the system. Priority: Medium.

Stimulus/Response Sequences:

9. Administrator authenticates.
10. Administrator views configured sources.
11. Administrator adds or changes source details.

12. System validates and stores the configuration.

Functional Requirements:

FR-29: The system shall allow an authorized administrator to manage the configured trusted sources.

FR-30: The system shall prevent unauthorized users from changing trusted-source configuration.

5.10 Logging and Review

Description and Priority: Maintains records needed to understand verification activity and review flagged results. Priority: Medium.

Stimulus/Response Sequences:

13. System records verification activity.
14. Administrator/reviewer accesses permitted records.
15. Flagged or uncertain results can be reviewed.
16. Logs can be used for debugging and project evaluation.

Functional Requirements:

6. Other Nonfunctional Requirements

6.1 Performance Requirements

- The system should begin processing a valid request without unnecessary delay.
- The response time should be acceptable for interactive chatbot use under normal network and source conditions.
- The system should avoid repeated retrieval of identical evidence where caching is implemented.
- Performance should degrade gracefully when an external source is slow or unavailable.
- The exact quantitative response-time target is TBD and should be finalized after the selected implementation and test environment are fixed.

6.2 Safety Requirements

- The system shall not present an uncertain verification result as a confirmed fact.
- When trusted evidence is unavailable, the system should indicate that verification could not be completed.
- The module should preserve uncertainty instead of silently modifying factual content without an indication where practical.
- The system should not be used as the sole safety-critical decision mechanism without domain-specific validation.

6.3 Security Requirements

- Only authorized users shall access administrative functions.
- Trusted-source configuration shall be protected from unauthorized modification.
- API requests should use authentication when exposed beyond a controlled local environment.
- HTTPS should be used for network communication containing protected information.
- Logs should avoid storing unnecessary sensitive user information.

- Stored records should be protected against unauthorized access or modification.

6.4 Software Quality Attributes

- Correctness: verification results should be based on the evidence actually retrieved.
- Reliability: failures in external sources should be handled without falsely reporting successful verification.
- Usability: results and evidence should be easy for a normal user to understand.
- Maintainability: claim extraction, retrieval, and verification components should be separable so they can be updated independently.
- Interoperability: the API should allow integration with different AI applications.
- Testability: each major verification stage should be testable independently.
- Portability: the software should be deployable in commonly used development/server environments.
- Traceability: verification results should be linked to the claims and evidence used to produce them.

6.5 Business Rules

- Only configured trusted sources shall be used as trusted evidence sources.
- Administrative source-management functions shall be restricted to authorized administrators.
- A claim shall not be marked supported without sufficient evidence according to the implemented verification rule.
- Contradicted and uncertain results shall remain distinguishable from supported results.

Reviewer access shall be limited to the records required for evaluation and review